

Mathematics A

Unit 2 • Chapter OL-T22 Classified

Cross-session • chapter-organised candidate

8 classified items • 27 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T22 • 8 items • 27 marks

Chapter	Items	Marks	Starts
01. OL-T22 Rearranging Formulas	8	27	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Rearranging Formulas

Unit 2 • 27 marks • 8 item(s)

June 2025 | 4MA1-2H Q16 | 4 marks

Single squared/powered subject

January 2022 | 4MA1-1H Q1 | 2 marks

Single squared/powered subject

January 2020 | 4MA1-2H Q16 | 4 marks

Subject in the denominator

January 2021 | 4MA1-2H Q14 | 4 marks

Subject in the denominator

June 2021 | 4MA1-1H Q5 | 2 marks

Linear add/multiply formula

November 2021 | 4MA1-1H Q17 | 3 marks

Subject in the denominator

May-Nov 2020 | 2H Q15 | 4 marks

Linear add/multiply formula

November 2025 | 1H Q18 | 4 marks

Subject in a product with several factors

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

16 Make t the subject of the formula $c = \frac{t^2 + 3}{7 - 8t^2}$

(Total for Question 16 is 4 marks)

Worked Solution 27Make t the subject**UNIT 2**
4 marks**Clear the denominator**

$$c(7 - 8t^2) = t^2 + 3 \Rightarrow 7c - 8ct^2 = t^2 + 3.$$

Collect and factor

$$7c - 3 = (1 + 8c)t^2.$$

Take square roots

$$t^2 = \frac{7c - 3}{1 + 8c} \Rightarrow t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}.$$

Final answer

$$t = \pm \sqrt{\frac{7c - 3}{1 + 8c}}$$

Original question 01

4MA1-1H Q1 Q1(d)

ORIGINAL QUESTION**2 marks**

(d) Make t the subject of $c = t^3 - 8v$

(2)

Worked solution 01

Chapter: Rearranging Formulas

MODULAR UNIT 2

2 marks

Q1(d) – Chapter: Rearranging Formulas

Family: Undo a single power or root in a formula • Variant: Single squared/powered subject

Method $c = t^3 - 8v$ gives $t^3 = c + 8v$.*Why: Add $8v$ to both sides.***Method** $t = \text{cube_root}(c + 8v)$.*Why: Take the real cube root of both sides.***Final answer****t:** $t = \sqrt[3]{c + 8v}$

16 Make x the subject of $y = \sqrt{\frac{x+1}{x-4}}$

Worked solution

January 2020 | Question 16 | 4 marks

Family: Rearrange a subject inside one fraction • Variant: Subject in the denominator

Method / working

1. Square the equation: $y^2 = (x+1)/(x-4)$.
 2. Multiply through by $x-4$ to remove the fraction.
 3. Expand and collect x terms: $x(y^2-1) = 4y^2+1$.
 4. Divide by y^2-1 : $x = (4y^2+1)/(y^2-1)$.
- Final answer: $x = (4y^2+1)/(y^2-1)$

(b) Make c the subject of the formula $p = \sqrt{\frac{ac + 8}{3 + c}}$

(4)

Worked solution

January 2021 | Question 14 | 4 marks

Family: Rearrange a subject inside one fraction • Variant: Subject in the denominator

Method / working

1. Square: $p^2 = (ac+8)/(c+3)$.
 2. Multiply through: $p^2c+3p^2=ac+8$.
 3. Collect c terms: $c(p^2-a)=8-3p^2$.
 4. Hence $c=(8-3p^2)/(p^2-a)$.
- Final answer: $c=(8-3p^2)/(p^2-a)$

June 2021 | 4MA1-1H | Question 5 | 2 marks

Chapter OL-T22: Rearranging Formulas

(b) Make t the subject of the formula $p = at - d$

(2)

Worked solution

June 2021 | Question 5 | 2 marks

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

Method / working

1. Add d to both sides of $p = at - d$: $p+d = at$.
2. Divide by a to make t the subject.

Final answer: $t = (p+d)/a$

(b) Make f the subject of $w = \frac{2f + 3}{8 - f}$

(3)

Worked solution

November 2021 | Question 17 | 3 marks

Family: Rearrange a subject inside one fraction • Variant: Subject in the denominator

Method / working

1. (b)

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

$$2. w = \frac{2f+3}{8-f}$$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

$$3. w(8-f) = 2f+3$$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

$$4. 8w-3 = f(w+2)$$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

$$5. f = \frac{8w-3}{w+2}$$

Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.

Final answer: $(6y+5)(y-1)$, $f = \frac{8w-3}{w+2}$, $4(x-1)^2+3$

15 Make x the subject of $y = \frac{5 - 2x}{x + 3}$

(Total for Question 15 is 4 marks)

Answer reference | May-Nov 2020 | Question 15

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

15	$xy + 3y = 5 - 2x$ oe			M1	multiplying both sides by $(x + 3)$ and expanding the brackets correctly
	e.g. $xy + 2x = 5 - 3y$			M1	ft dep on 2 terms on left and $(5 - 2x)$ on right, for collecting all x terms on one side and non- x terms on the other side
	eg $x(y + 2) = 5 - 3y$			M1	ft, dep on 2 terms in x , for factorising for x
		$x = \frac{5-3y}{2+y}$	4	A1	oe allow $\frac{5-3y}{2+y}$ as answer so long as previously seen $x = \frac{5-3y}{2+y}$
					Total 4 marks

Worked solution

May-Nov 2020 | Question 15 | 4 marks

Family: Isolate a subject through inverse operations • Variant: Linear add/multiply formula

Official final mark-scheme pages are embedded immediately after this question.

18 Make y the subject of $p = \sqrt{\frac{y + w}{3y - t}}$

(Total for Question 18 is 4 marks)

Worked solution

November 2025 | Question 18 | 4 marks

Family: Isolate a subject through inverse operations • Variant: Subject in a product with several factors

Solution

$$p = \sqrt{(y+w)/(3y-t)}$$

$$p^2 = (y+w)/(3y-t)$$

$$p^2(3y-t) = y+w$$

$$3p^2y - p^2t = y+w$$

$$y(3p^2-1) = w+p^2t$$

$$\text{Answer: } y = (w+p^2t)/(3p^2-1)$$